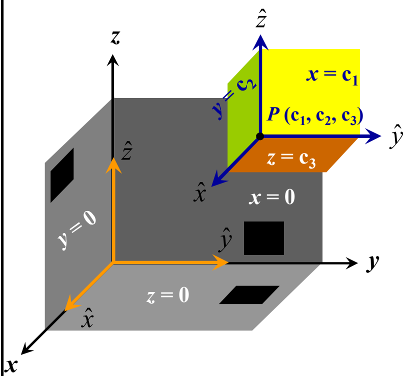
Curvilinear Coordinate Systems

Cartesian Coordinate Systems (x, y, z)



Length element:

$$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$$

Area element: $dx\,dy\,\hat{z}$

$dy\,dz\,\hat{x}$

$dz\,dx\,\hat{y}$

$$d\vec{a} = dy\,dz\,\hat{x} + dz\,dx\,\hat{y} + dx\,dy\,\hat{z}$$

Volume element:

$$d\tau = dx\,dy\,dz$$

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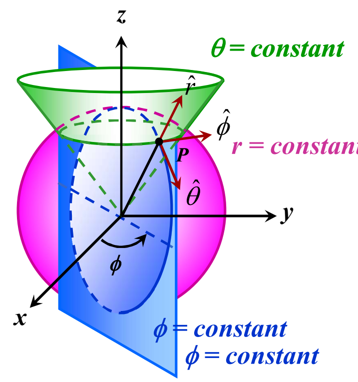
Curvilinear Coordinate Systems

Spherical Polar Coordinate Systems (r, θ, ϕ)



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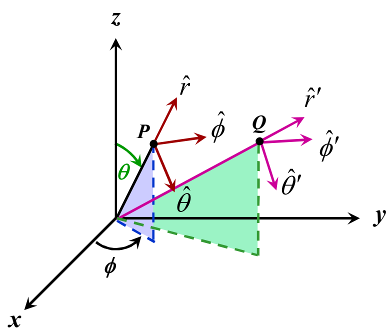
Curvilinear Coordinate Systems

Spherical Polar Coordinate Systems (r, θ, ϕ)

$r: 0 \rightarrow \infty$

$\theta: 0 \rightarrow \pi$

$\phi: 0 \rightarrow 2\pi$



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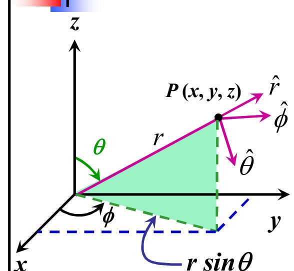
Spherical Polar Coordinate Systems



$$x = r\sin\theta\cos\phi = x(r,\theta,\phi)$$

$$y = r\sin\theta\sin\phi = y(r,\theta,\phi)$$

$$z = r\cos\theta = z(r,\theta,\phi)$$

$$r = (x^2 + y^2 + z^2)^{1/2} = r(x,y,z)$$

$$\sqrt{x^2 + y^2} = r\sin\theta(\cos^2\phi + \sin^2\phi)^{1/2}$$

$$\theta = \tan^{-1}\left(\sqrt{x^2+y^2}\Big/z\right) = \theta(x,y,z)$$

$$\varphi = \tan^{-1}\left(y/x\right) = \phi(x,y,z)$$

$$\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}$$

$$\frac{\partial\vec{r}}{\partial r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z} \longrightarrow \hat{r}$$

$$\frac{\partial\vec{r}}{\partial\theta} = r\cos\theta\cos\phi\,\hat{x} + r\cos\theta\sin\phi\,\hat{y} - r\sin\theta\,\hat{z} \longrightarrow \frac{1}{r}\frac{\partial\vec{r}}{\partial\theta} = \hat{\theta}$$

$$\frac{\partial\vec{r}}{\partial\phi} = -r\sin\theta\sin\phi\,\hat{x} + r\sin\theta\cos\phi\,\hat{y} \longrightarrow \frac{1}{r\sin\theta}\frac{\partial\vec{r}}{\partial\phi} = \hat{\phi}$$

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Spherical Polar Coordinate Systems

$$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$$\begin{pmatrix}\hat{r}\\ \hat{\theta}\\ \hat{\varphi}\end{pmatrix} = \begin{pmatrix}\sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta\\ \cos\theta\cos\varphi & \cos\theta\sin\varphi & -\sin\theta\\ -\sin\varphi & \cos\varphi & 0\end{pmatrix}\begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix}$$

$$\begin{pmatrix}\hat{r}\\ \hat{\theta}\\ \hat{\phi}\end{pmatrix} = \mathbf{A}\begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix} \longrightarrow \begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix}\hat{r}\\ \hat{\theta}\\ \hat{\phi}\end{pmatrix}$$

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Spherical Polar Coordinate Systems

$$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$$\frac{\partial\hat{r}}{\partial r} = 0$$

$$\frac{\partial\hat{r}}{\partial\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z} = \hat{\theta} \neq 0$$

$$\frac{\partial\hat{r}}{\partial\phi} = -\sin\theta\sin\phi\,\hat{x} + \sin\theta\cos\phi\,\hat{y} = \hat{\phi} \neq 0$$

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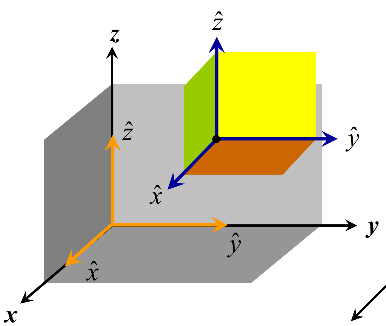
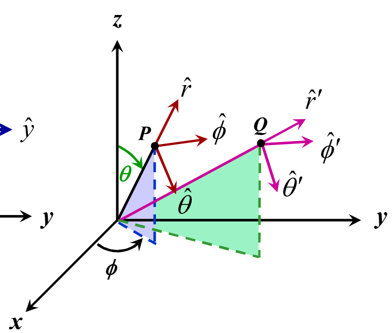
Curvilinear Coordinate Systems





Be careful while integrating/differentiating $(\hat{r}, \hat{\theta}, \hat{\phi})$

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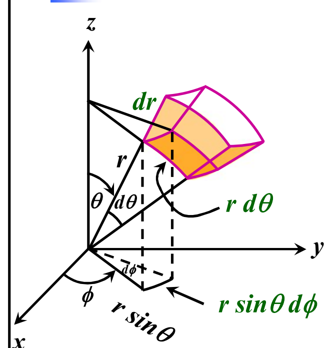
Spherical Polar Coordinate Systems



Length element: $d\vec{l} = ?$

$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ in CCS

$d\vec{l} = dl_r\,\hat{r} + dl_\theta\,\hat{\theta} + dl_\phi\,\hat{\phi}$

$= \cancel{dr\,\hat{r} + d\theta\,\hat{\theta} + d\phi\,\hat{\phi}}$

$$\boxed{d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}}$$

Area element: $r dr\,d\theta\;\hat{\phi}$

$r\sin\theta\,dr\,d\phi\;\hat{\theta}$

$r^2\sin\theta\,d\theta\,d\phi\;\hat{r}$

$$\boxed{d\vec{a} = r^2\sin\theta\,d\theta\,d\phi\;\hat{r} + r\sin\theta\,dr\,d\phi\,\hat{\theta} + r\,dr\,d\theta\,\hat{\phi}}$$

Volume element: $\boxed{d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi}$

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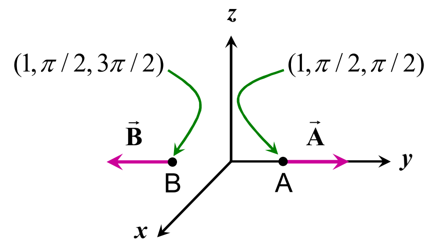
Understanding



$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$

<u>In CCS</u>	<u>In SPCS</u>
$\vec{\mathbf{A}} = \hat{y}$	$\vec{\mathbf{A}} = \hat{r}$
$\vec{\mathbf{B}} = -\hat{y}$	$\vec{\mathbf{B}} = \hat{r}$
$\vec{\mathbf{A}} + \vec{\mathbf{B}} = \mathbf{0}$	$\vec{\mathbf{A}} + \vec{\mathbf{B}} = 2\hat{r}$
$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = -1$	$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = 1$

?

$\hat{r}_A = \hat{y}$

$\hat{r}_B = -\hat{y}$

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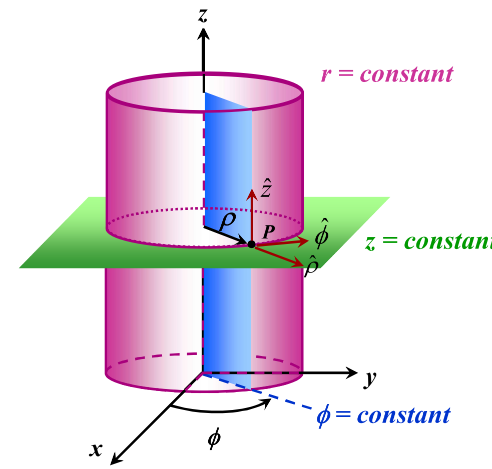
Curvilinear Coordinate Systems

Cylindrical Polar Coordinate Systems (ρ, ϕ, z)



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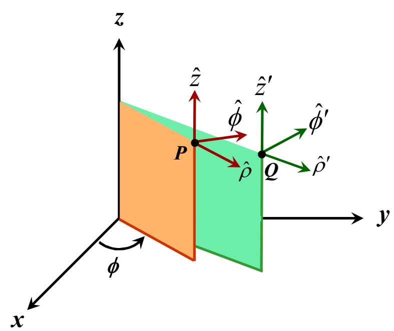
Curvilinear Coordinate Systems

Cylindrical Polar Coordinate Systems (ρ, ϕ, z)

$\rho : 0 \rightarrow \infty$

$\phi : 0 \rightarrow 2\pi$

$z : -\infty \rightarrow +\infty$



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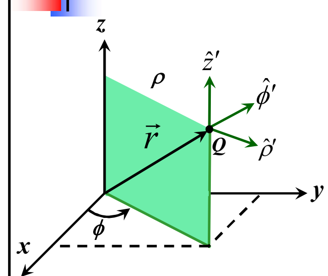
Cylindrical Polar Coordinate Systems



$x = \rho\cos\varphi \;\; = x(\rho,\varphi,z)$

$y = \rho\sin\varphi \;\; = y(\rho,\varphi,z)$

$z = z \;\; = z(\rho,\varphi,z)$

$\rho = (x^2 + y^2)^{1/2} = \rho\ (x,y,z)$

$\phi = \tan^{-1}(y/x) = \varphi\ (x,y,z)$

$z = z \;\; = z\ (x,y,z)$

$\vec{r} = \rho\cos\varphi\,\hat{x} + \rho\sin\varphi\,\hat{y} + z\,\hat{z}$

$\dfrac{\partial\vec{r}}{\partial\rho} = \cos\phi\,\hat{x} + \sin\phi\,\hat{y} \longrightarrow \hat{\rho}$

$\dfrac{\partial\vec{r}}{\partial\phi} = -\rho\sin\varphi\,\hat{x} + \rho\cos\varphi\,\hat{y} \longrightarrow \dfrac{1}{\rho}\dfrac{\partial\vec{r}}{\partial\varphi} = \hat{\phi}$

$\dfrac{\partial\vec{r}}{\partial z} = \hat{z} \longrightarrow \hat{z}$

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Cylindrical Polar Coordinate Systems

$$\hat{\rho} = \cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}$$

$$\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\hat{z} = \hat{z}$$

$$\begin{pmatrix}\hat{\rho}\\ \hat{\varphi}\\ \hat{z}\end{pmatrix} = \begin{pmatrix}\cos\varphi & \sin\varphi & 0\\ -\sin\varphi & \cos\varphi & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix}$$

$$\begin{pmatrix}\hat{\rho}\\ \hat{\varphi}\\ \hat{z}\end{pmatrix} = \mathrm{A}\begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix} \longrightarrow \begin{pmatrix}\hat{x}\\ \hat{y}\\ \hat{z}\end{pmatrix} = \mathrm{A}^{-1}\begin{pmatrix}\hat{\rho}\\ \hat{\varphi}\\ \hat{z}\end{pmatrix}$$

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Cylindrical Polar Coordinate Systems

$$\hat{\rho} = \cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}$$

$$\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\hat{z} = \hat{z}$$

$$\frac{\partial\hat{\rho}}{\partial r} = 0 \qquad \frac{\partial\hat{\phi}}{\partial\rho} = 0$$

$$\frac{\partial\hat{\rho}}{\partial\varphi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y} \;= \hat{\phi} \neq 0 \qquad \frac{\partial\hat{\phi}}{\partial\phi} = -\cos\phi\,\hat{x} - \sin\phi\,\hat{y} \;= -\hat{r} \neq 0$$

$$\frac{\partial\hat{\rho}}{\partial z} = 0 \qquad \frac{\partial\hat{\phi}}{\partial z} = 0$$

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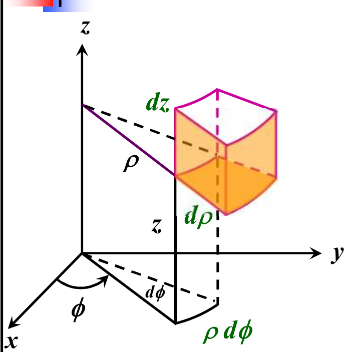
Cylindrical Polar Coordinate Systems



Length element: $d\vec{l} = ?$

$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ in CCS

$d\vec{l} = dl_{\rho}\,\hat{\rho} + dl_{\varphi}\,\hat{\varphi} + dl_{z}\,\hat{z}$

$\quad = \cancel{d\rho\,\hat{\rho} + d\varphi\,\hat{\varphi} + dz}\,\hat{z}$

$$\boxed{d\vec{l} = d\rho\,\hat{\rho} + \rho\,d\varphi\,\hat{\varphi} + dz\,\hat{z}}$$

Area element: $d\rho\,dz\;\hat{\varphi}$

$\rho\,d\rho\,d\varphi\;\hat{z}$

$\rho\,dz\,d\varphi\;\hat{\rho}$

$$\boxed{d\vec{a} = \rho\,dz\,d\varphi\;\hat{\rho} + d\rho\,dz\,\hat{\varphi} + \rho\,d\rho\,d\varphi\,\hat{z}}$$

Volume element: $\boxed{d\tau = \rho\,d\rho\,d\varphi\,dz}$

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Curvilinear Coordinate Systems: Summary

$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ $\qquad x \quad y \quad z$

$\quad dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$ $\qquad r \quad \theta \quad \phi$

$\quad d\rho\,\hat{\rho} + \rho\,d\varphi\,\hat{\varphi} + dz\,\hat{z}$ $\qquad \rho \quad \varphi \quad z$

$$d\vec{a} = \left.\begin{matrix} dy\,dz\,\hat{x}\\ dz\,dx\,\hat{y}\\ dx\,dy\,\hat{z}\end{matrix}\right\} \quad \left.\begin{matrix} r^2\sin\theta\,d\theta\,d\phi\;\hat{r}\\ r\sin\theta\,d\phi\,dr\;\hat{\theta}\\ r\,dr\,d\theta\;\hat{\phi}\end{matrix}\right\} \quad \left.\begin{matrix} \rho\,d\varphi\,dz\;\hat{\rho}\\ dz\,d\rho\;\hat{\varphi}\\ \rho\,d\rho\,d\varphi\;\hat{z}\end{matrix}\right\}$$

$d\tau = dx\,dy\,dz$

$\quad r^2\sin\theta\,dr\,d\theta\,d\phi$

$\quad r\,dr\,d\phi\,dz$

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Curvilinear Coordinate Systems: Summary

h_1 $\quad$ h_2 $\quad$ h_3	ξ_1	ξ_2	ξ_3		h_1	h_2	h_3
$d\vec{l} = 1\,dx\,\hat{x} + 1\,dy\,\hat{y} + 1\,dz\,\hat{z}$	x	y	z	$\rightarrow$	1	1	1
$1\,dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$	r	θ	ϕ	$\rightarrow$	1	r	$r\sin\theta$
$1\,d\rho\,\hat{\rho} + \rho\,d\varphi\,\hat{\varphi} + 1\,dz\,\hat{z}$	ρ	φ	z	$\rightarrow$	1	ρ	1

$$\boxed{d\vec{l} = h_1\,d\xi_1\,\hat{\xi}_1 + h_2\,d\xi_2\,\hat{\xi}_2 + h_3\,d\xi_3\,\hat{\xi}_3}$$

$$d\vec{a} = \left.\begin{matrix} dy\,dz\,\hat{x} \\ dz\,dx\,\hat{y} \\ dx\,dy\,\hat{z} \end{matrix}\right\} \quad \left.\begin{matrix} r^2\sin\theta\,d\theta\,d\phi\;\hat{r} \\ r\sin\theta\,d\phi\,dr\;\hat{\theta} \\ r\,dr\,d\theta\;\hat{\phi} \end{matrix}\right\} \quad \left.\begin{matrix} \rho\,d\varphi\,dz\;\hat{\rho} \\ dz\,d\rho\;\hat{\varphi} \\ \rho\,d\rho\,d\varphi\;\hat{z} \end{matrix}\right\}$$

$$\boxed{d\vec{a} = h_2h_3d\xi_2d\xi_3\,\hat{\xi}_1 + h_3h_1d\xi_3d\xi_1\,\hat{\xi}_2 + h_1h_2d\xi_1d\xi_2\,\hat{\xi}_3}$$

$$d\tau = dx\,dy\,dz$$
$$r^2\sin\theta\,dr\,d\theta\,d\phi$$
$$r\,dr\,d\phi\,dz$$

$$\boxed{d\tau = h_1h_2h_3d\xi_1d\xi_2d\xi_3}$$

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Gradient in General Curvilinear Coordinate System

Consider a scalar function $\psi(\xi_1, \xi_2, \xi_3)$

$$\vec{\nabla}\psi = \alpha_1\hat{\xi}_1 + \alpha_2\hat{\xi}_2 + \alpha_3\hat{\xi}_3$$

$$d\psi = \vec{\nabla}\psi\cdot d\vec{l} \qquad d\vec{l} = h_1\,d\xi_1\,\hat{\xi}_1 + h_2\,d\xi_2\,\hat{\xi}_2 + h_3\,d\xi_3\,\hat{\xi}_3$$

$$= h_1\alpha_1 d\xi_1 + h_2\alpha_2 d\xi_2 + h_3\alpha_3 d\xi_3$$

$$d\psi = \frac{\partial\psi}{\partial\xi_1}d\xi_1 + \frac{\partial\psi}{\partial\xi_2}d\xi_2 + \frac{\partial\psi}{\partial\xi_3}d\xi_3$$

$$\alpha_1 = \frac{1}{h_1}\frac{\partial\psi}{\partial\xi_1} \qquad \alpha_2 = \frac{1}{h_2}\frac{\partial\psi}{\partial\xi_2} \qquad \alpha_3 = \frac{1}{h_3}\frac{\partial\psi}{\partial\xi_3}$$

$$\vec{\nabla}\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial\xi_1}\hat{\xi}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial\xi_2}\hat{\xi}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial\xi_3}\hat{\xi}_3$$

$$\boxed{\vec{\nabla} \equiv \hat{\xi}_1\frac{1}{h_1}\frac{\partial}{\partial\xi_1} + \hat{\xi}_2\frac{1}{h_2}\frac{\partial}{\partial\xi_2} + \hat{\xi}_3\frac{1}{h_3}\frac{\partial}{\partial\xi_3}}$$

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Divergence in General Curvilinear Coordinate System

$$\vec{\nabla}\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial\xi_1}\hat{\xi}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial\xi_2}\hat{\xi}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial\xi_3}\hat{\xi}_3$$

Consider a scalar function $\psi = \xi_1 \quad \psi = \xi_2 \quad \psi = \xi_3$

$$\vec{\nabla}\xi_1 = \frac{\hat{\xi}_1}{h_1} \qquad \vec{\nabla}\xi_2 = \frac{\hat{\xi}_2}{h_2} \qquad \vec{\nabla}\xi_3 = \frac{\hat{\xi}_3}{h_3}$$

$$\vec{\nabla}\xi_1\times\vec{\nabla}\xi_2 = \frac{\hat{\xi}_1\times\hat{\xi}_2}{h_1h_2} = \frac{\hat{\xi}_3}{h_1h_2}$$

$$\hat{\xi}_3 = h_1h_2\vec{\nabla}\xi_1\times\vec{\nabla}\xi_2$$

$$\hat{\xi}_1 = h_2h_3\vec{\nabla}\xi_2\times\vec{\nabla}\xi_3$$

$$\hat{\xi}_2 = h_3h_1\vec{\nabla}\xi_3\times\vec{\nabla}\xi_1$$

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Divergence in General Curvilinear Coordinate System

$$\hat{\xi}_1 = h_2h_3\vec{\nabla}\xi_2\times\vec{\nabla}\xi_3 \qquad \hat{\xi}_2 = h_3h_1\vec{\nabla}\xi_3\times\vec{\nabla}\xi_1 \qquad \hat{\xi}_3 = h_1h_2\vec{\nabla}\xi_1\times\vec{\nabla}\xi_2$$

$$\vec{V} = V_1\hat{\xi}_1 + V_2\hat{\xi}_2 + V_3\hat{\xi}_3$$

$$\vec{\nabla}\cdot\vec{V} = \vec{\nabla}\cdot(V_1\hat{\xi}_1 + V_2\hat{\xi}_2 + V_3\hat{\xi}_3) \qquad \vec{\nabla}\cdot(f\,\vec{V})$$

$$\vec{\nabla}\cdot(V_1\hat{\xi}_1) = \vec{\nabla}\cdot(V_1h_2h_3\vec{\nabla}\xi_2\times\vec{\nabla}\xi_3) \qquad \vec{\nabla}\cdot(\vec{A}\times\vec{B})$$

$$= (V_1h_2h_3)\vec{\nabla}\cdot(\vec{\nabla}\xi_2\times\vec{\nabla}\xi_3) + \vec{\nabla}(V_1h_2h_3)\cdot(\vec{\nabla}\xi_2\times\vec{\nabla}\xi_3)$$

$$= (V_1h_2h_3)[\vec{\nabla}\xi_2\cdot(\cancel{\vec{\nabla}\times\vec{\nabla}\xi_2}) - \vec{\nabla}\xi_2\cdot(\cancel{\vec{\nabla}\times\vec{\nabla}\xi_3})] + \vec{\nabla}(V_1h_2h_3)\cdot\frac{\hat{\xi}_1}{h_2h_3}$$

$$= \vec{\nabla}(V_1h_2h_3)\cdot\frac{\hat{\xi}_1}{h_2h_3}$$

$$= \left[\frac{1}{h_1}\frac{\partial(V_1h_2h_3)}{\partial\xi_1}\hat{\xi}_1 + (\ )\hat{\xi}_2 + (\ )\hat{\xi}_3\right]\cdot\frac{\hat{\xi}_1}{h_2h_3} \qquad = \frac{1}{h_1h_2h_3}\frac{\partial(V_1h_2h_3)}{\partial\xi_1}$$

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Divergence in General Curvilinear Coordinate System

$$\vec{\nabla}\cdot\vec{V} = \underline{\vec{\nabla}\cdot(V_1\hat{\xi}_1} + \underline{V_2\hat{\xi}_2} + \underline{V_3\hat{\xi}_3})$$

$$\vec{\nabla}\cdot(V_1\hat{\xi}_1) = \frac{1}{h_1h_2h_3}\frac{\partial(V_1h_2h_3)}{\partial\xi_1}$$

$$\vec{\nabla}\cdot(V_2\hat{\xi}_2) = \frac{1}{h_1h_2h_3}\frac{\partial(V_2h_3h_1)}{\partial\xi_2}$$

$$\vec{\nabla}\cdot(V_3\hat{\xi}_3) = \frac{1}{h_1h_2h_3}\frac{\partial(V_3h_1h_2)}{\partial\xi_3}$$

$$\boxed{\vec{\nabla}\cdot\vec{V} = \frac{1}{h_1h_2h_3}\left[\frac{\partial(V_1h_2h_3)}{\partial\xi_1} + \frac{\partial(V_2h_3h_1)}{\partial\xi_2} + \frac{\partial(V_3h_1h_2)}{\partial\xi_3}\right]}$$

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Curl in General Curvilinear Coordinate System

$$\vec{\nabla}\times\vec{V} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{\xi}_1 & h_2\hat{\xi}_2 & h_3\hat{\xi}_3 \\ \partial/\partial\xi_1 & \partial/\partial\xi_2 & \partial/\partial\xi_3 \\ h_1V_1 & h_2V_2 & h_3V_3 \end{vmatrix}$$

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Laplacian in General Curvilinear Coordinate System

$$\nabla^2\psi = \vec{\nabla}\cdot\vec{\nabla}\psi = \vec{\nabla}\cdot\vec{V} = \frac{1}{h_1h_2h_3}\left[\frac{\partial(V_1h_2h_3)}{\partial\xi_1} + \frac{\partial(V_2h_3h_1)}{\partial\xi_2} + \frac{\partial(V_3h_1h_2)}{\partial\xi_3}\right]$$

$$= \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial\xi_1}\left(\frac{1}{h_1}\frac{\partial\psi}{\partial\xi_1}h_2h_3\right) + \frac{\partial}{\partial\xi_2}\left(\frac{1}{h_2}\frac{\partial\psi}{\partial\xi_2}h_3h_1\right) + \frac{\partial}{\partial\xi_3}\left(\frac{1}{h_3}\frac{\partial\psi}{\partial\xi_3}h_1h_2\right)\right]$$

$$\boxed{\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial\xi_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial\xi_1}\right) + \frac{\partial}{\partial\xi_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial\xi_2}\right) + \frac{\partial}{\partial\xi_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial\xi_3}\right)\right]}$$

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Summary

$$\vec{\nabla}\psi = \frac{1}{h_1}\frac{\partial\psi}{\partial\xi_1}\hat{\xi}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial\xi_2}\hat{\xi}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial\xi_3}\hat{\xi}_3$$

$$\vec{\nabla}\cdot\vec{V} = \frac{1}{h_1h_2h_3}\left[\frac{\partial(V_1h_2h_3)}{\partial\xi_1} + \frac{\partial(V_2h_3h_1)}{\partial\xi_2} + \frac{\partial(V_3h_1h_2)}{\partial\xi_3}\right]$$

$$\vec{\nabla}\times\vec{V} = \frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{\xi}_1 & h_2\hat{\xi}_2 & h_3\hat{\xi}_3 \\ \partial/\partial\xi_1 & \partial/\partial\xi_2 & \partial/\partial\xi_3 \\ h_1V_1 & h_2V_2 & h_3V_3 \end{vmatrix}$$

	h_1	h_2	h_3
CCS	1	1	1
SPCS	1	r	$r\sin\theta$
CPCS	1	ρ	1

$$\nabla^2\psi = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial\xi_1}\left(\frac{h_2h_3}{h_1}\frac{\partial\psi}{\partial\xi_1}\right) + \frac{\partial}{\partial\xi_2}\left(\frac{h_3h_1}{h_2}\frac{\partial\psi}{\partial\xi_2}\right) + \frac{\partial}{\partial\xi_3}\left(\frac{h_1h_2}{h_3}\frac{\partial\psi}{\partial\xi_3}\right)\right]$$

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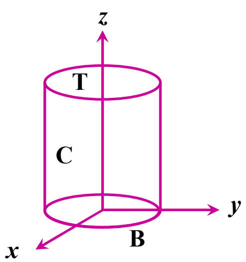
Example

Determine the flux of a vector field $\vec{F}(r) = 10\,e^{-2z}(r\,\hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \le z \le 1$.

Confirm the result using the divergence theorem.

$$\text{Flux} = \oint \vec{F}\cdot d\vec{a} = \int_T \vec{F}\cdot d\vec{a} + \int_C \vec{F}\cdot d\vec{a} + \int_B \vec{F}\cdot d\vec{a}$$

For **T**, $r: 0 \to 1,\ \phi: 0 \to 2\pi,\ z = 1$



$$d\vec{a} = h_1 h_2 d\xi_1 d\xi_2\,\hat{\xi}_3 = r\,dr d\phi\,\hat{z}$$

$$\vec{F}\cdot d\vec{a} = 10\,e^{-2z} dr d\phi \;= 10\,e^{-2} dr d\phi$$

$$\text{Flux from } \mathbf{T} = \int_{r=0}^{1}\int_{\phi=0}^{2\pi} 10\,e^{-2} r\,dr d\phi$$

$$= 10\,e^{-2}\int_{r=0}^{1} r\,dr \int_{\phi=0}^{2\pi} d\phi = \boxed{10\,\pi\,e^{-2}}$$

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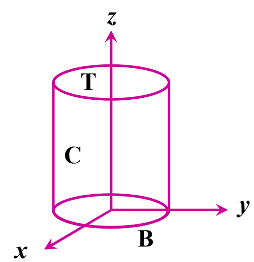
Example

Determine the flux of a vector field $\vec{F}(r) = 10\,e^{-2z}(r\,\hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \le z \le 1$.

Confirm the result using the divergence theorem.

$$\text{Flux} = \oint \vec{F}\cdot d\vec{a} = \int_T \vec{F}\cdot d\vec{a} + \int_C \vec{F}\cdot d\vec{a} + \int_B \vec{F}\cdot d\vec{a}$$

For **B**, $r: 0 \to 1,\ \phi: 0 \to 2\pi,\ z = 0$



$$d\vec{a} = -\,r\,dr d\phi\,\hat{z}$$

$$\vec{F}\cdot d\vec{a} = -10\,e^{-2z} dr d\phi = -10\,dr d\phi$$

$$\text{Flux from } \mathbf{B} = -\int_{r=0}^{1}\int_{\phi=0}^{2\pi} 10\,r\,dr d\phi$$

$$= -10\int_{r=0}^{1} r\,dr \int_{\phi=0}^{2\pi} d\phi = \boxed{-10\,\pi}$$

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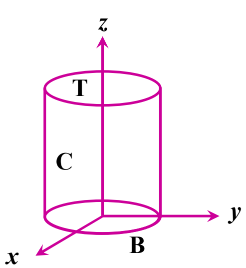
Example

Determine the flux of a vector field $\vec{F}(r) = 10\,e^{-2z}(r\,\hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \le z \le 1$.

Confirm the result using the divergence theorem.

$$\text{Flux} = \oint \vec{F}\cdot d\vec{a} = \int_T \vec{F}\cdot d\vec{a} + \int_C \vec{F}\cdot d\vec{a} + \int_B \vec{F}\cdot d\vec{a}$$

For **C**, $r = 1,\ \phi: 0 \to 2\pi,\ z: 0 \to 1$



$$d\vec{a} = h_2 h_3 d\xi_2 d\xi_3\,\hat{\xi}_1 = r\,d\phi\,dz\,\hat{r}$$

$$\vec{F}\cdot d\vec{a} = 10\,e^{-2z} r^2 d\phi\,dz \;= 10\,e^{-2z} d\phi\,dz$$

$$\text{Flux from } \mathbf{C} = -\int_{\phi=0}^{2\pi}\int_{z=0}^{1} 10\,e^{-2z}\,d\phi\,dz$$

$$= \boxed{10\,\pi\,(1 - e^{-2})}$$

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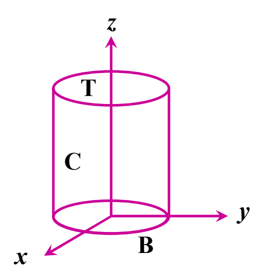
Example

Determine the flux of a vector field $\vec{F}(r) = 10\,e^{-2z}(r\,\hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \le z \le 1$.

Confirm the result using the divergence theorem.

$$\text{Flux} = \oint \vec{F}\cdot d\vec{a} = \int_T \vec{F}\cdot d\vec{a} + \int_C \vec{F}\cdot d\vec{a} + \int_B \vec{F}\cdot d\vec{a}$$



Flux from **T** $= 10\,\pi\,e^{-2}$

Flux from **B** $= -10\,\pi$

Flux from **C** $= 10\,\pi\,(1 - e^{-2})$

$\boxed{\text{Total Flux} = 0}$

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Example

Determine the flux of a vector field $\vec{F}(r) = 10\, e^{-2z}(r\,\hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \le z \le 1$.

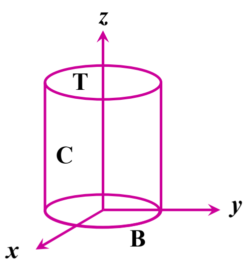
Confirm the result using the divergence theorem.

$$\vec{\nabla}\cdot\vec{V} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial(V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial(V_3 h_1 h_2)}{\partial \xi_3}\right]$$

$$= \frac{1}{r}\left[\frac{\partial(10\, e^{-2z} r r)}{\partial r} + \frac{\partial(0)}{\partial \phi} + \frac{\partial(10 e^{-2z} r)}{\partial z}\right]$$

$$= \frac{1}{r}\left[10 e^{-2z}\frac{\partial r^2}{\partial r} + 0 + 10\, r\frac{\partial e^{-2z}}{\partial z}\right]$$

$$= \frac{1}{r}\left[10 e^{-2z} 2r + 0 + 10\, r(-2e^{-2z})\right] = 0$$



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